RES-OPT — Design Optimisation

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1 Domain Purpose and Scope

RES-OPT is deferred under the current 2026-07-19 scope decision. The active Research work compares fixed-rigid barrier classes in the Kamaishi and Sendai Tohoku corridors using hydrodynamic metrics; it does not run surrogate optimisation, unrestricted inverse design or final design selection. Future optimisation may consume validated barrier-class outputs after data handling, hydrodynamic load extraction and the V1–V2–V3 validation hierarchy are accepted.

2 Domain Deliverables

3 RES-OPT-PRB — Optimisation Problem and Design-Space Definition

3.1 Purpose and Scope

This Deliverable records the optimisation problem as deferred. The active project defines barrier-class comparison cases and metrics, not an optimisation search space. Future optimisation must start from validated hydrodynamic outputs and accepted geometry descriptors.

3.2 Research Questions

3.3 Terminology and Definitions

3.4 Literature Evidence

3.5 Governing Relations and Quantitative Definitions

3.6 Decision Variables

Future decision variables may be controlled geometry descriptors admitted by RES-GEO-SPEC, such as class, magnitude and placement, subject to constructability and numerical-feasibility constraints. They are not active optimisation variables in the current scope.

3.7 Objective Functions

3.8 Constraints

3.9 Sampling Strategy

3.10 Optimisation Method

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3.20 Requirements and Handoffs

3.21 Deliverable Completion Record

4 RES-OPT-OBJ — Objective Functions and Performance Measures

4.1 Purpose and Scope

This Deliverable records future objective-function candidates but does not activate optimisation. In the current project phase, run-up, overtopping, transmission/reflection, pressure, force, impulse, moment and energy are comparison metrics for fixed-rigid barrier classes, not optimisation objectives. Structural response, adequacy, damage classification and surrogate uncertainty are deferred.

4.2 Research Questions

4.3 Terminology and Definitions

4.4 Literature Evidence

4.5 Governing Relations and Quantitative Definitions

4.6 Decision Variables

4.7 Objective Functions

Future objective components may include reducing peak force, overturning moment, overtopping, transmitted energy or downstream hazard while preserving constructability. Displacement, stress, permanent deformation, damage probability and surrogate uncertainty are excluded until structural and ML extensions are activated and validated.

4.8 Constraints

4.9 Sampling Strategy

4.10 Optimisation Method

4.11 Computational Budget

4.12 Termination and Selection Criteria

4.13 Candidate Approaches

4.14 Critical Comparison

4.15 Assumptions and Applicability

4.16 Limitations and Failure Conditions

4.17 Project-Specific Conclusions

For the active Studentship, these quantities are reported as hydrodynamic comparison metrics only. No objective aggregation, final design ranking or optimisation claim shall be made from unvalidated outputs.

4.18 Selected Approach and Rationale

The selected active approach is metric reporting, not optimisation. Future objective functions may be defined after RES-VER-MET accepts the relevant hydrodynamic quantities.

4.19 Unresolved Decisions

4.20 Requirements and Handoffs

4.21 Deliverable Completion Record

5 RES-OPT-CST — Constraints, Feasibility and Design Admissibility

5.1 Purpose and Scope

This Deliverable records future optimisation constraints as deferred. The active project uses RES-GEO-CST feasibility checks for fixed-rigid barrier-class comparison; it does not define an admissible optimisation space or reinforcement-intervention campaign.

5.2 Research Questions

5.3 Terminology and Definitions

5.4 Literature Evidence

5.5 Governing Relations and Quantitative Definitions

5.6 Decision Variables

5.7 Objective Functions

5.8 Constraints

Future constraints may include geometry validity, constructability, material limits, foundation limits, meshability, numerical feasibility, validated-domain limits and direct-simulation confirmation requirements. In the active scope, only geometry validity, terrain/corridor fit, meshability and unresolved-physics flags are required for barrier-class comparison.

5.9 Sampling Strategy

5.10 Optimisation Method

5.11 Computational Budget

5.12 Termination and Selection Criteria

5.13 Candidate Approaches

5.14 Critical Comparison

5.15 Assumptions and Applicability

5.16 Limitations and Failure Conditions

5.17 Project-Specific Conclusions

5.18 Selected Approach and Rationale

5.19 Unresolved Decisions

5.20 Requirements and Handoffs

5.21 Deliverable Completion Record

6 RES-OPT-DOE — Design of Experiments and Simulation-Sampling Strategy

6.1 Purpose and Scope

This Deliverable is deferred beyond the active barrier-class comparison. The current scope may define a small reproducible set of wall, obstacle, dissipating and array cases, but it does not define an optimisation DOE or ML training campaign.

6.2 Research Questions

6.3 Terminology and Definitions

6.4 Literature Evidence

6.5 Governing Relations and Quantitative Definitions

6.6 Decision Variables

6.7 Objective Functions

6.8 Constraints

6.9 Sampling Strategy

Future sampling may cover baseline cases, bounded intervention options, controlled event-condition variations and independent final test simulations. High-uncertainty surrogate regions, failure-boundary regions and ML-driven active sampling are deferred until validated outputs exist.

6.10 Optimisation Method

6.11 Computational Budget

6.12 Termination and Selection Criteria

6.13 Candidate Approaches

6.14 Critical Comparison

6.15 Assumptions and Applicability

6.16 Limitations and Failure Conditions

6.17 Project-Specific Conclusions

6.18 Selected Approach and Rationale

6.19 Unresolved Decisions

6.20 Requirements and Handoffs

6.21 Deliverable Completion Record

7 RES-OPT-ALG — Optimisation Algorithm Assessment and Selection

7.1 Purpose and Scope

7.2 Research Questions

7.3 Terminology and Definitions

7.4 Literature Evidence

7.5 Governing Relations and Quantitative Definitions

7.6 Decision Variables

7.7 Objective Functions

7.8 Constraints

7.9 Sampling Strategy

7.10 Optimisation Method

fixed-barrier metric extraction and validation planning.

10.2 Research Questions

10.3 Terminology and Definitions

10.4 Literature Evidence

10.5 Governing Relations and Quantitative Definitions

10.6 Decision Variables

10.7 Objective Functions

10.8 Constraints

10.9 Sampling Strategy

10.10 Optimisation Method

10.11 Computational Budget

No model-training or optimisation budget is accepted in the active scope. Computational budget records shall include solver fidelity, convergence requirements, quality labels, expected reruns and validation cases. Direct FSI confirmation triggered by surrogate uncertainty is deferred.

10.12 Termination and Selection Criteria

10.13 Candidate Approaches

10.14 Critical Comparison

10.15 Assumptions and Applicability

10.16 Limitations and Failure Conditions

10.17 Project-Specific Conclusions

10.18 Selected Approach and Rationale

10.19 Unresolved Decisions

10.20 Requirements and Handoffs

10.21 Deliverable Completion Record

11 RES-OPT-SPEC — Integrated Optimisation Specification

11.1 Purpose and Scope

This Deliverable records that optimisation is deferred for the active Tohoku hydrodynamic scope. It preserves existing WBS IDs while requiring any future optimisation to consume validated barrier-class metrics rather than driving the present Research work.

11.2 Research Questions

11.3 Terminology and Definitions

11.4 Literature Evidence

11.5 Governing Relations and Quantitative Definitions

11.6 Decision Variables

11.7 Objective Functions

11.8 Constraints

11.9 Sampling Strategy

11.10 Optimisation Method

11.11 Computational Budget

11.12 Termination and Selection Criteria

11.13 Candidate Approaches

11.14 Critical Comparison

11.15 Assumptions and Applicability

11.16 Limitations and Failure Conditions

11.17 Project-Specific Conclusions

11.18 Selected Approach and Rationale

The selected active approach is no optimisation execution and no final optimisation algorithm selection. Future controlled screening may be reconsidered only after fixed-rigid hydrodynamic metrics, uncertainty flags and validation status are available.

11.19 Unresolved Decisions

11.20 Requirements and Handoffs

Software shall not implement optimisation logic from this Research handoff. It shall preserve structured barrier-class metrics and provenance so a later optimisation extension can consume validated outputs.

11.21 Deliverable Completion Record